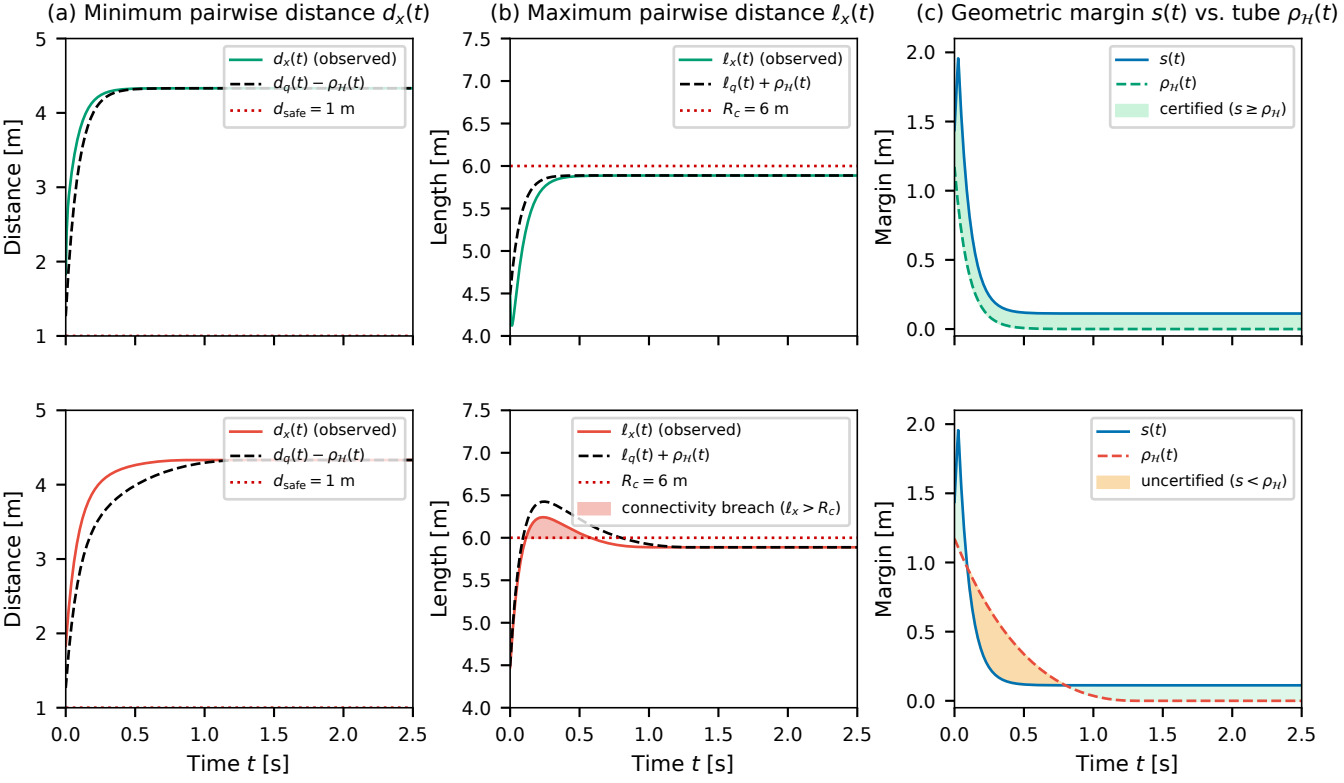

(a) Minimum pairwise distance $d_x(t)$
(b) Maximum pairwise distance $\ell_x(t)$
(c) Geometric margin $s(t)$ vs. tube $\rho_H(t)$
Distance [m]
Length [m]
Margin [m]
Time t [s]
$d_x(t)$ (observed)
$d_q(t) - \rho_H(t)$
$d_{\text{safe}} = 1$ m
$\ell_x(t)$ (observed)
$\ell_q(t) + \rho_H(t)$
$R_c = 6$ m
$s(t)$
$\rho_H(t)$
certified ($s \geq \rho_H$)
connectivity breach ($\ell_x > R_c$)
uncertified ($s < \rho_H$)